PROJECT BRIEF · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 12: Capstone: Algorithm App

Module 4 | 90 minutes | Capstone project | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Your mission
Build an app around a classic algorithm (search, sort, cipher or recursion): pseudocode first, the algorithm in a procedure, input and validation, clear output, step counts for several inputs, and a Python version of the algorithm.

Checklist

- [ ] Pseudocode
- [ ] Procedure
- [ ] Nested loops or recursion
- [ ] Input and validation
- [ ] Clear output
- [ ] Step counts
- [ ] Python version

Milestones

1. Pseudocode approved — done on: _____
2. Algorithm working — done on: _____
3. App around it — done on: _____
4. Python and measurements — done on: _____
5. Tested, submitted, presented — done on: _____

Plan it

Sketch your stage and write what each sprite will do.

How it will be marked

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Algorithm	Incorrect	Works for some inputs	Correct for all tested inputs	Correct and efficient, edge cases handled
Design	One script	Partly modular	Algorithm in a procedure	Clean, well-named procedures
Efficiency	Not measured	One count	Counts for several inputs	Explains growth (e.g. n^2)
Python	None	Fragments	Mostly correct translation	Accurate, idiomatic Python
Presentation	None	Demo only	Demo and explanation	Clear walkthrough of algorithm, blocks and Python

Want more?

- Visualise the algorithm with the pen
- Compare two algorithms
- Let users choose the algorithm

Two stars and a wish

Ask a friend to test your project.

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